

	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 May 2016	First Release	Tan Liang Kai

	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

Content

Conveyor motor removal.....	3
Conveyor motor replacement.....	5
Belt Alignment Verification.....	9

	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

** This procedure is applicable for conveyor motor on fixed rail & adjustable rail.

Conveyor motor removal

1. Open V810 > Services > Subsystem Status & Control > Initialize Panel Handling > Toggle Clamps into "Open" state.

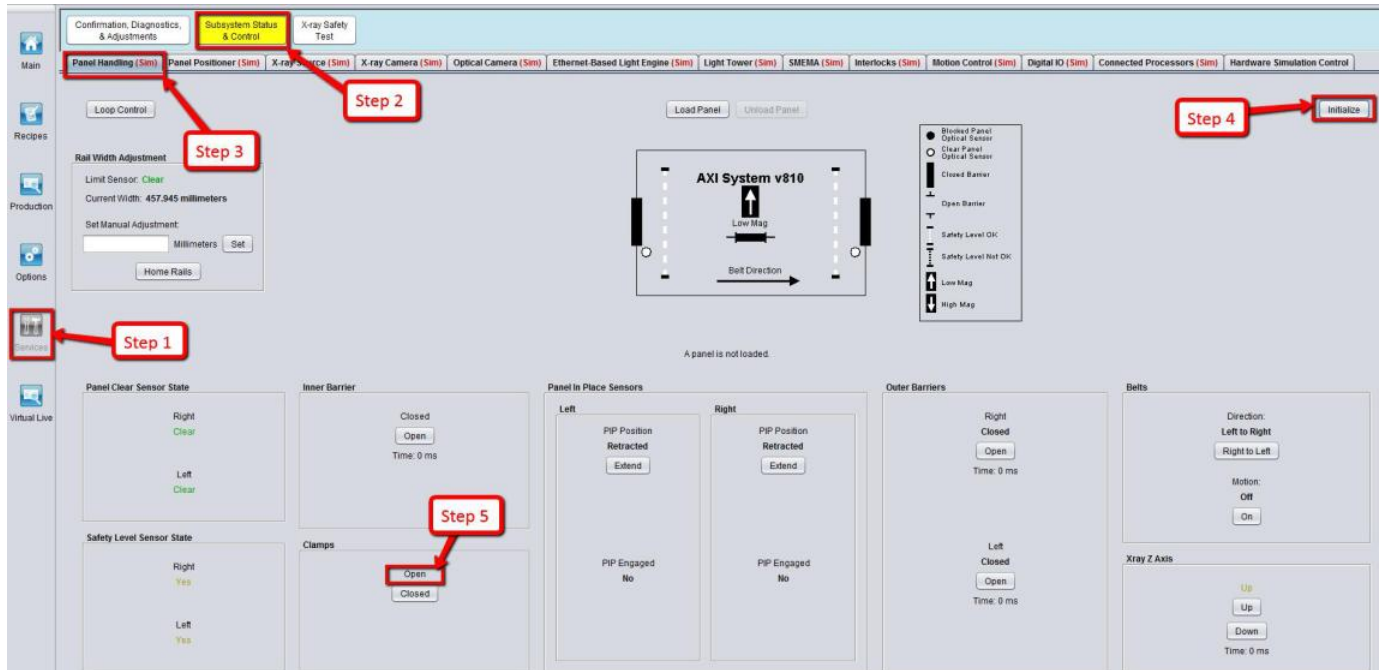


Figure 1 Panel Handling initialize

2. Close X-ray source.
3. Open Front Access Door.
4. Identify Defective Conveyor Motor Location by refer to below Figure 2.

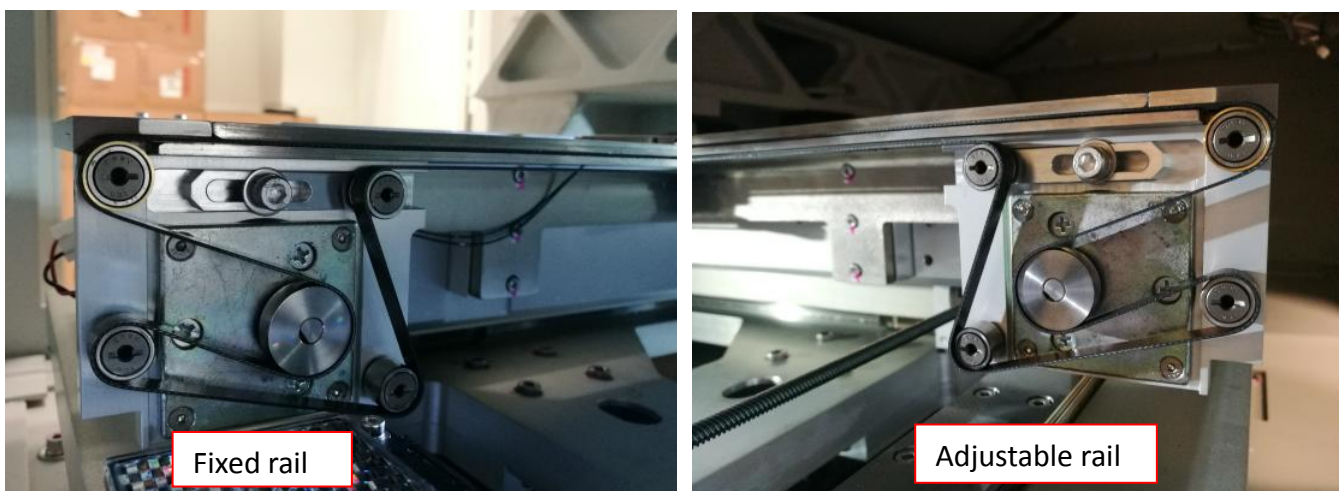


Figure 2 Conveyor Motor Location

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

5. Loosen Tension Pulley Screw (M5 x 20) & remove conveyor belt as shown in Figure 3.

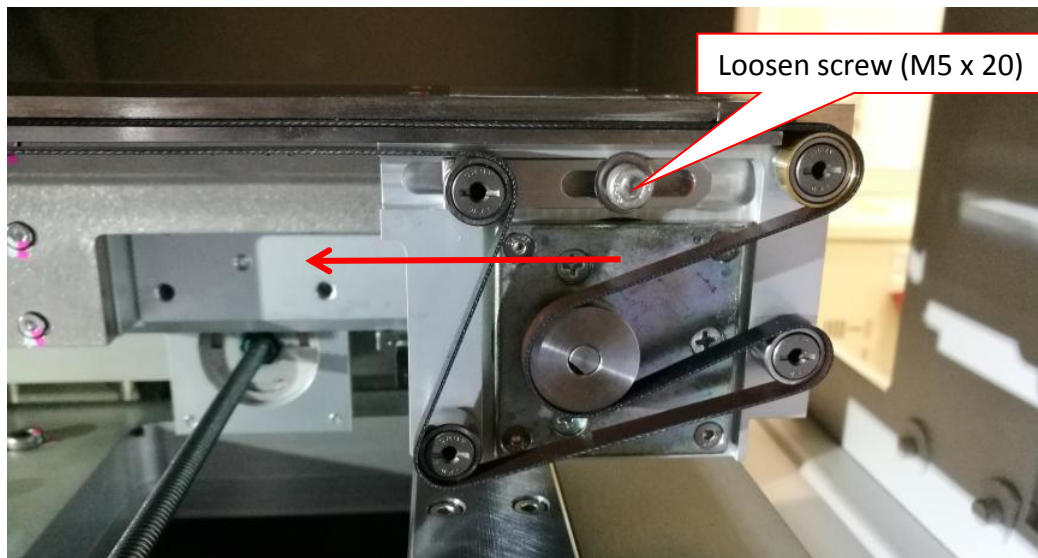


Figure 3 Conveyor Belt Removal

6. Unplug Conveyor Motor Connector as shown in Figure 4.

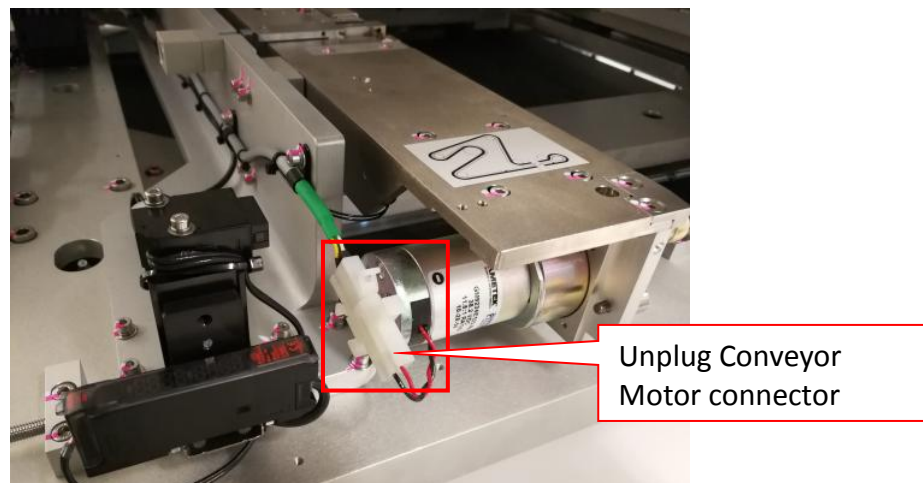


Figure 4 Conveyor Motor Connector

7. Remove 4 x screws (M3 x 10) & Conveyor Motor as shown in Figure 5.

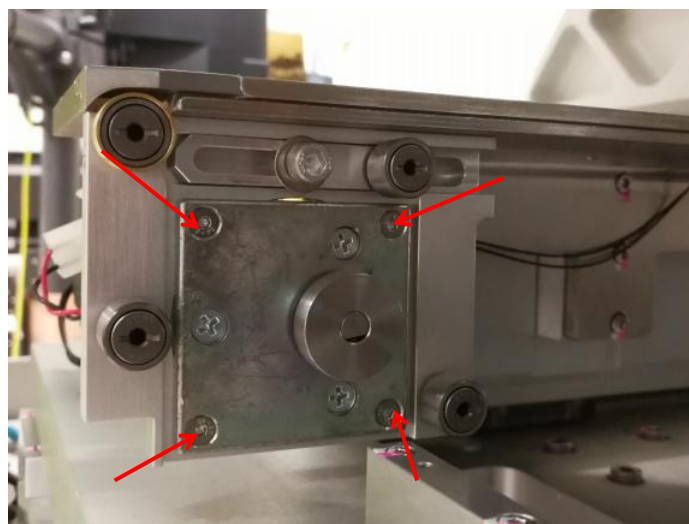


Figure 5 Conveyor Motor Removal

ViTrox 	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

8. Loosen Set Screw and gently remove Belt Pulley as shown in Figure 6.

Remarks : Belt Pulley & Set Screws require to re-use upon replacement

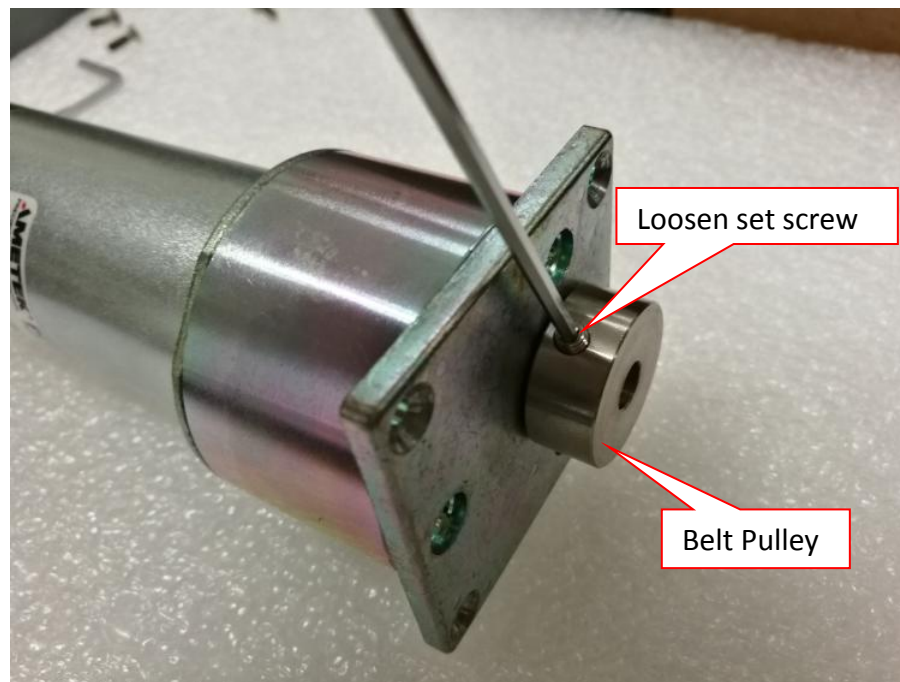


Figure 6 Belt Pulley Removal

Conveyor motor replacement

1. Prepare 1 PCS of 2670-0033 (MOTOR, 24VDC)
2. Assemble Belt pulley to Conveyor motor as shown in Figure 7.

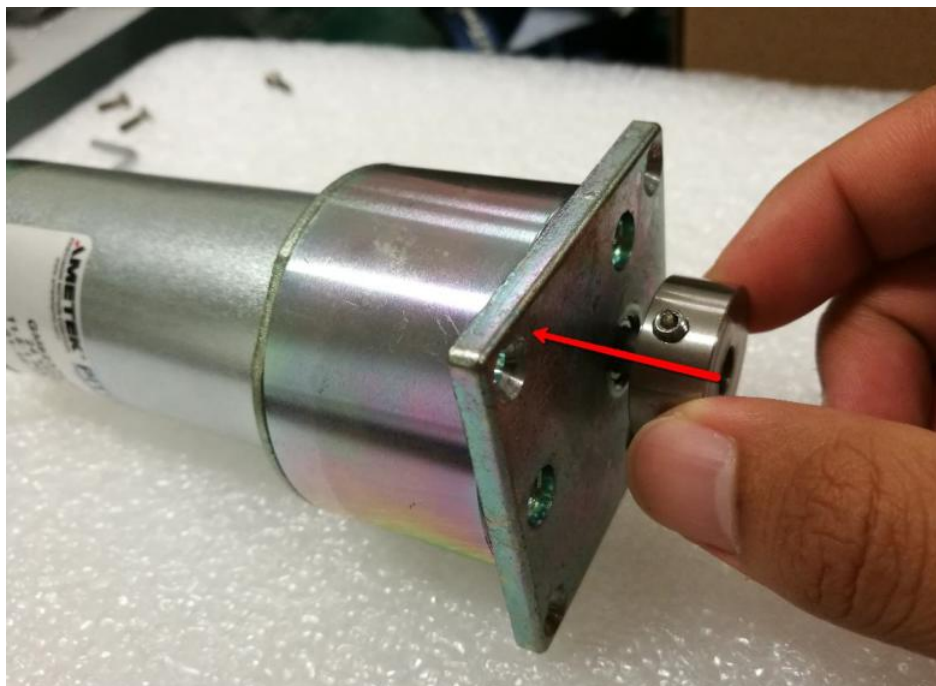


Figure 7 Belt Pulley Assembly

	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

2. Select 0.25 mm Filler Gauge & gently insert between Belt Pulley and Conveyor Motor as shown in Figure 8



Figure 8 Belt Pulley Assembly

3. While maintain Fillet Gauge position, tighten Set Screw as shown in Figure 9.

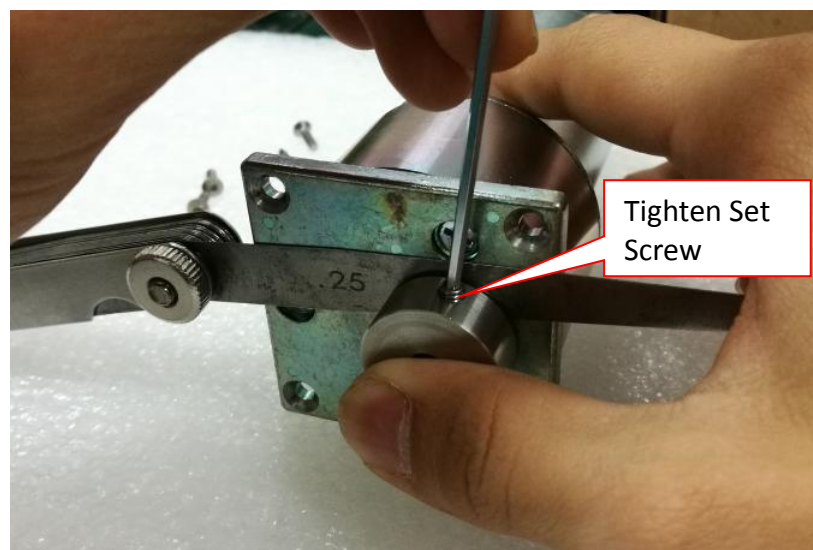


Figure 9 Belt Pulley Assembly

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

4. Assemble Conveyor motor & tighten 4 x screws (M3 x 10) as shown in Figure 10.

Remark: Caution on motor orientation.

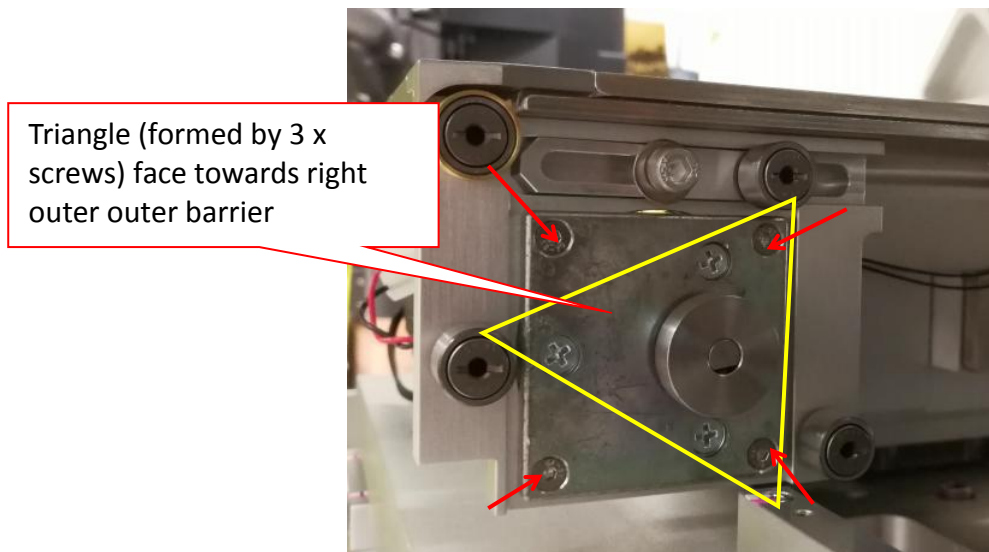


Figure 10 Conveyor Motor Assembly

5. Plug Conveyor Connector as shown in Figure 11.

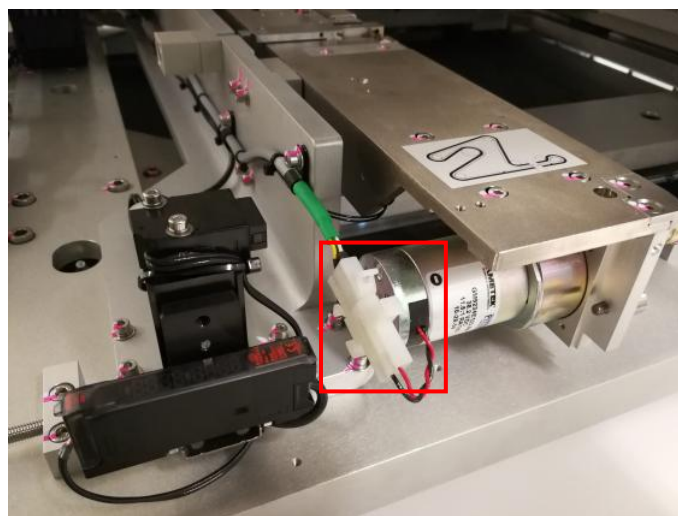


Figure 11 Conveyor Motor Connector

6. Assemble Conveyor Belt by refer to below routing. May refer to Figure 12.

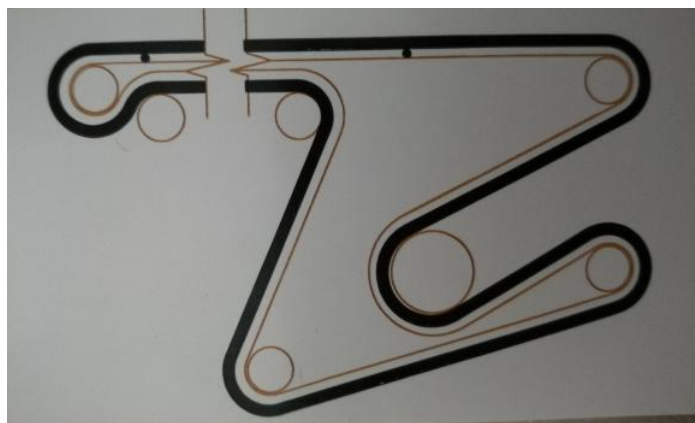


Figure 12 Conveyor belt routing

	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

7. May refer to below Figure 13 & 14 for Fix & Adj Rail Belt Routing

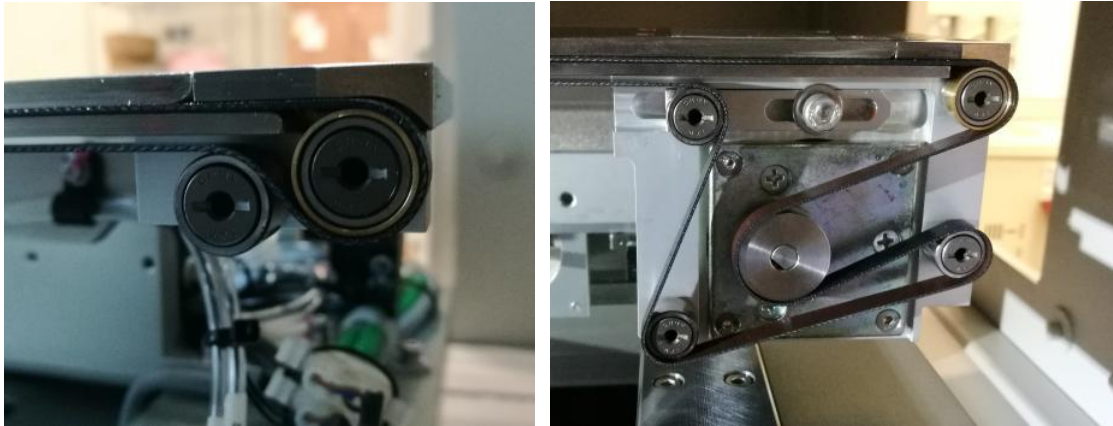


Figure 13 Adjustable Rail Belt Routing



Figure 14 Fixed Rail Belt Routing

8. Loosen the Belt Tension Pulley Block screw. Gently adjust Belt Tension Pulley Block is within 3/4 adjustment scale. Then, tighten the Belt Tension Pulley Block Screw. May refer to Figure 15 & 16.

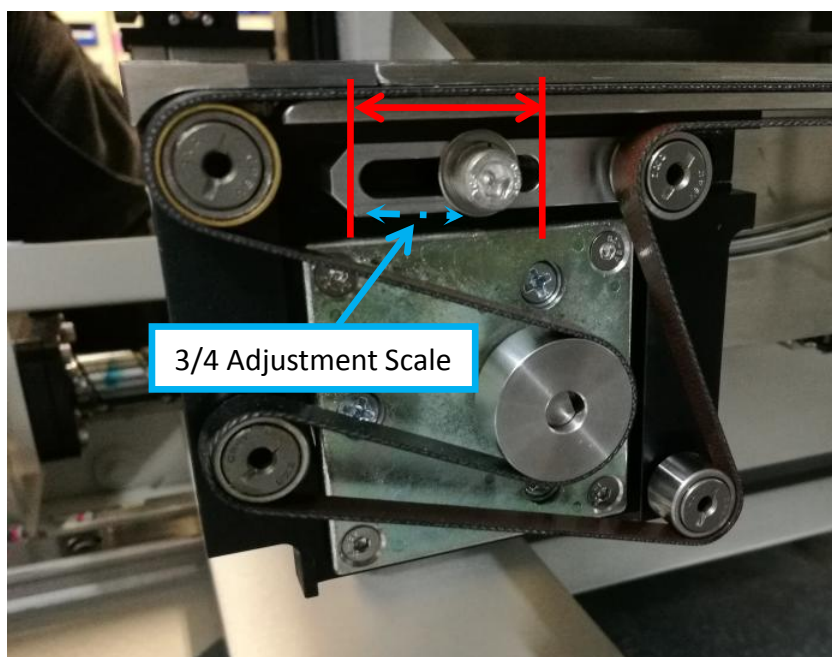


Figure 15 Fixed rail mount pulley

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

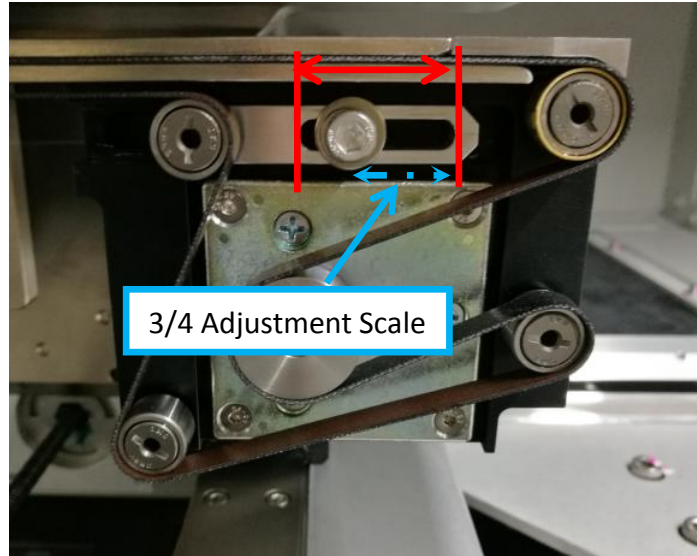


Figure 16 Adjustable rail mount pulley

Belt Alignment Verification

1. Turn on Conveyor Motor at Digital IO Tab (Stage belt Motor ON).

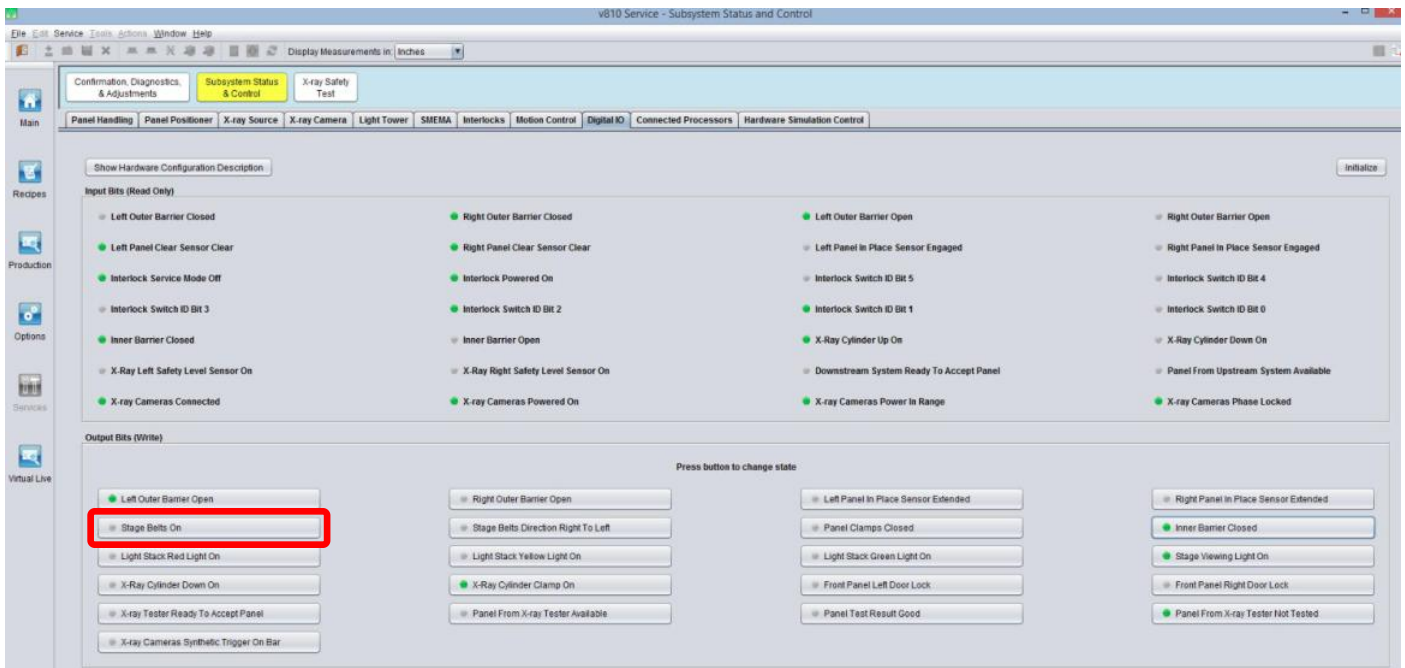


Figure 17 Belt Motor Toggle

	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

2. Visual check Conveyor Belt Running condition



Figure 18 Belt Flow (NG)

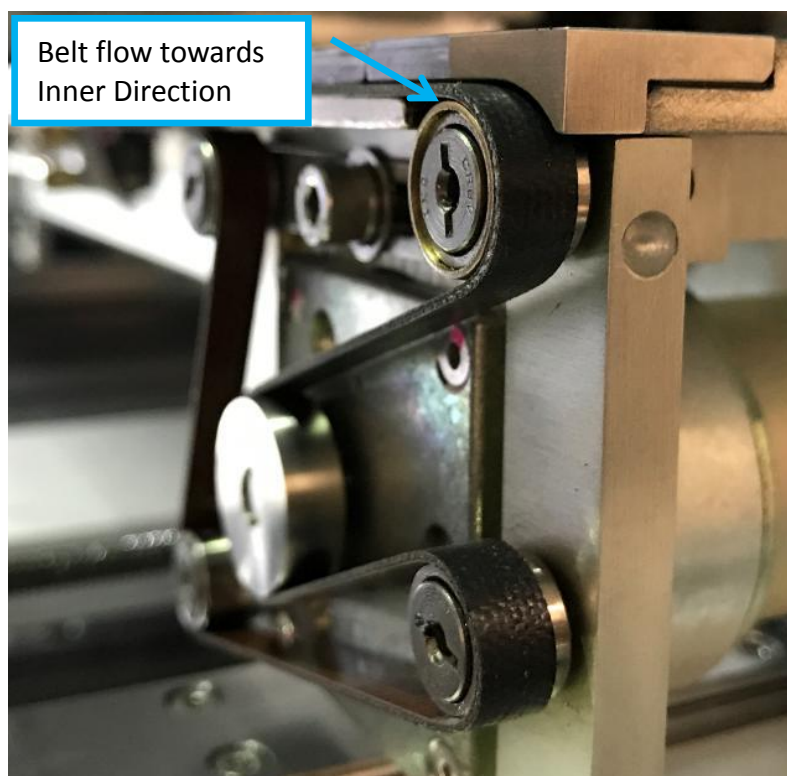


Figure 19 Belt Flow (Good)

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

3. Turn off the Belt Motor Motion at Digital IO Tab
4. Perform pulley alignment as below steps
 - a) Loosen 4 x Screws at Mount Pulley assembly as below Figure 20

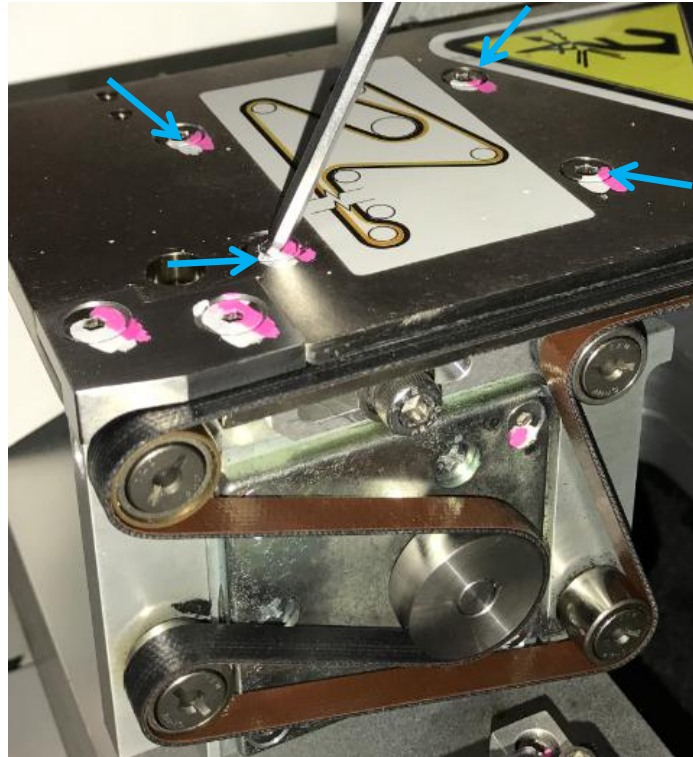


Figure 20 Adj Rail Mount Pulley Assembly

- b) Place L-ruler & Align Mount Pulley assembly as below Figure 21. Align Mount Pulley perpendicular with Adj Rail. Next, Tighten 4 x Screws

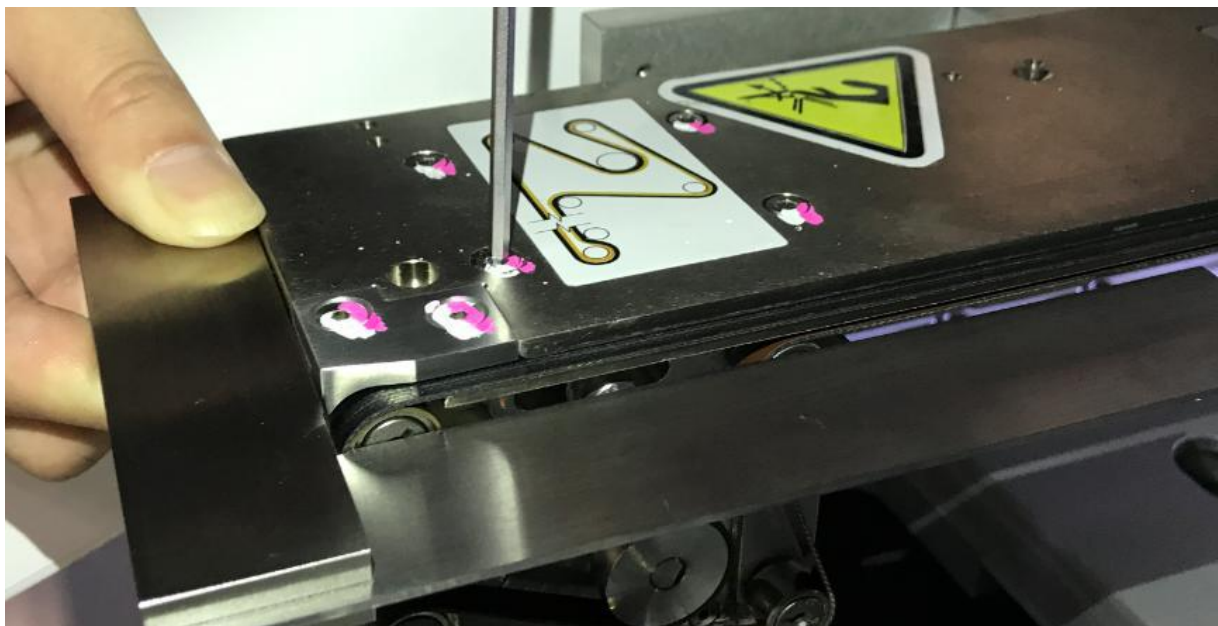


Figure 21 Adj Rail Mount Pulley Alignment

ViTrox 	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

- c) Toggle “Stage belt Motor ON “at Digital IO tab.
- d) Slightly loosen End Pulley assembly as below Figure 22

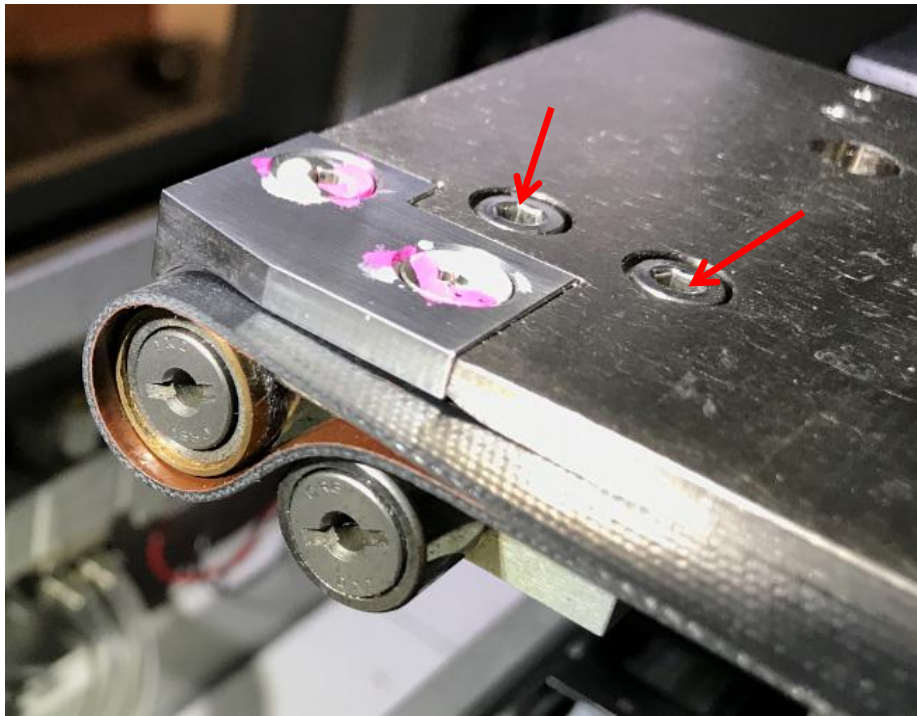


Figure 22 End Pulley Assembly

- e) Insert Allen Key to End Pulley as below Figure 23

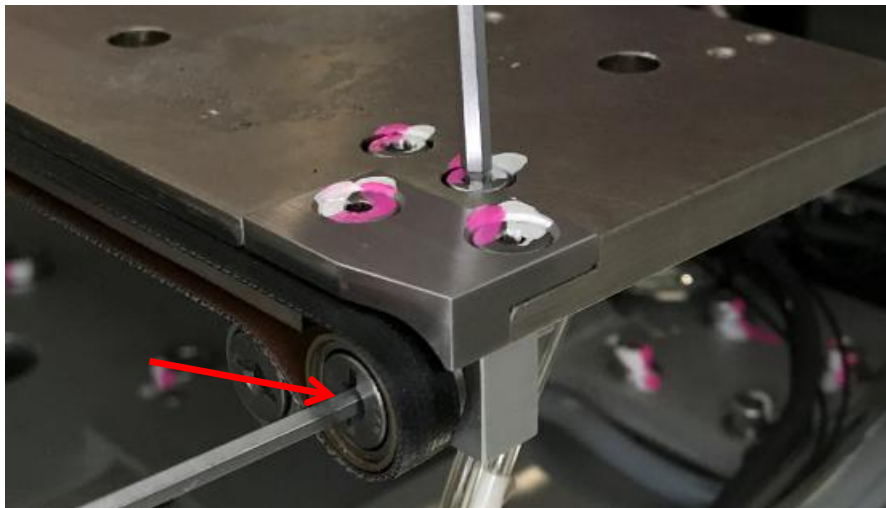


Figure 23 End Pulley Assembly

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 STD Conveyor Motor Removal & Replacement Procedure	Effective Date	30 May 2016

- f) Align the End Pulley until Belt Flow towards Inner Direction as below Figure 24. Tighten 2 x Screws after desire alignment achieve

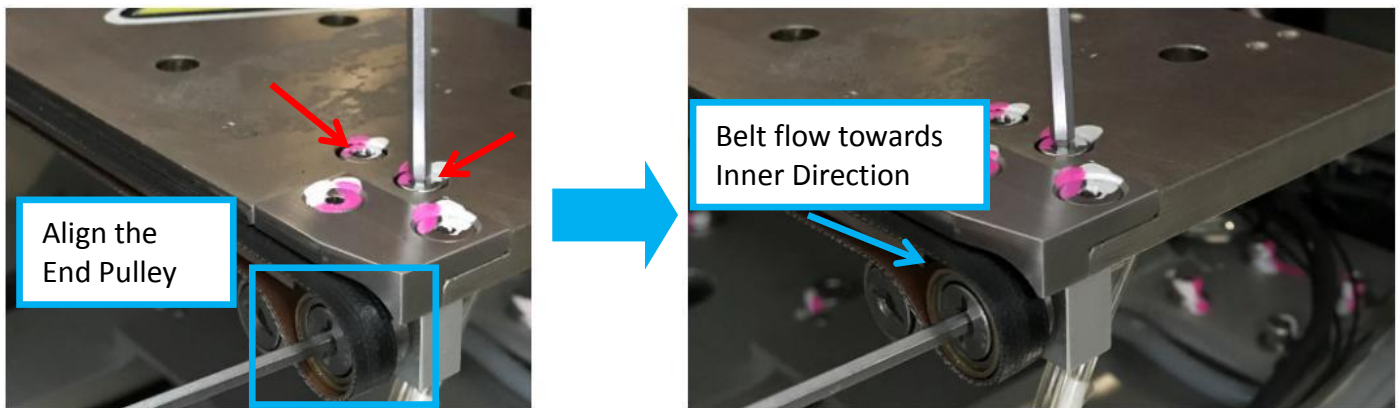


Figure 24 End Pulley Alignment

5. Toggle “Stage Belts Direction Right to Left “ while “Stage Belts On” remain ON.

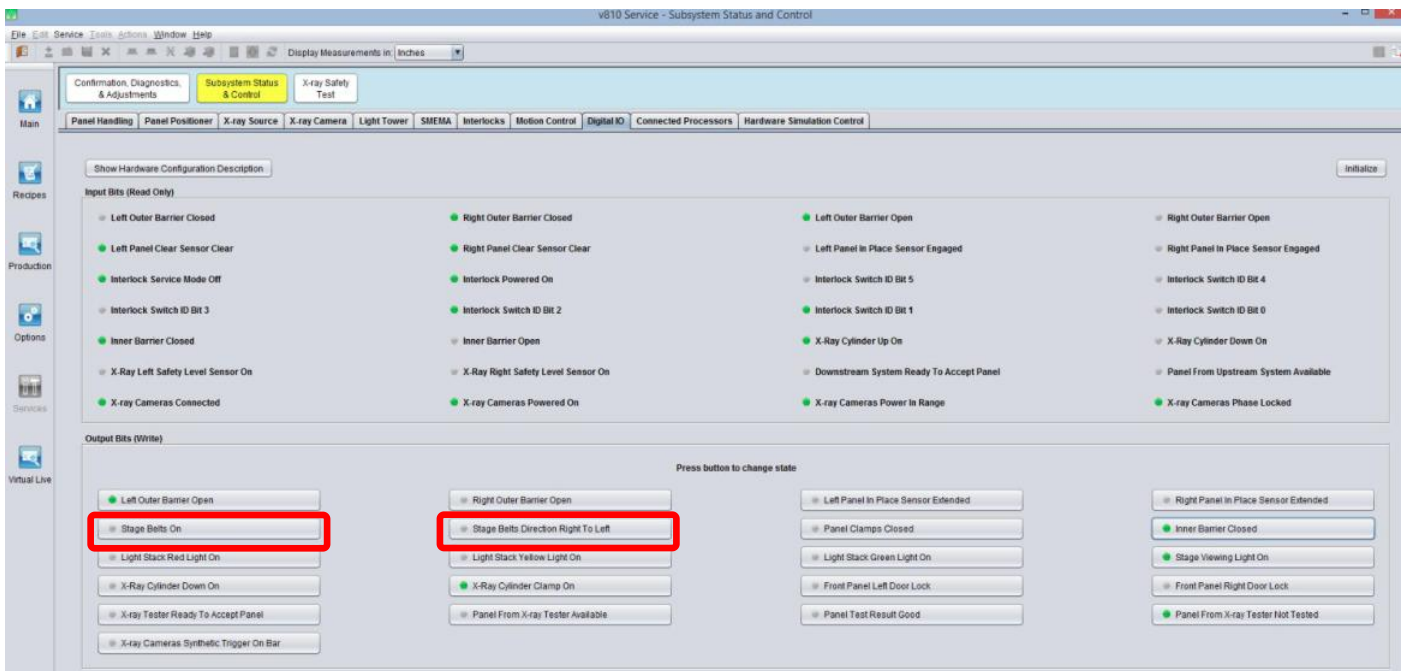


Figure 25 Digital IO (Belt Direction Switch)

6. Visual check Conveyor Belt Running Condition.
7. Repeat 2~6 if Conveyor Belt Running Condition fail
8. Toggle Off “Stage Belt On” at Digital IO tab
9. Toggle Off “ Stage Belt Direction Right to Left” at Digital IO tab
10. Remove all tools.Close all access door
11. Start machine as usual